

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think>

Number of vacuum cleaners sold at the green house = $1/3$ of the total

Let the total number of vacuum cleaners be x . Then, $1/3 * x$ were sold at the green house .

Number of vacuum cleaners left after selling at the green house = $x - 1/3 * x = 2/3 * x$

At the red house , 2 more vacuum cleaners were sold , so the number of vacuum cleaners left after selling at the red house = $2/3 * x - 2$

At the orange house , half of the remaining vacuum cleaners were sold , so the number of vacuum cleaners left after selling at the orange house = $(2/3 * x - 2) / 2 = (1/3 * x - 1)$

We are given that Melanie has 5 vacuum cleaners left , so $(1/3 * x - 1) = 5$

Solving for x , we get $1/3 * x = 6$, $x = 18$

Therefore , Melanie started with 18 vacuum cleaners .

</think>< answer >

18 </answer>